

Trainings ▾

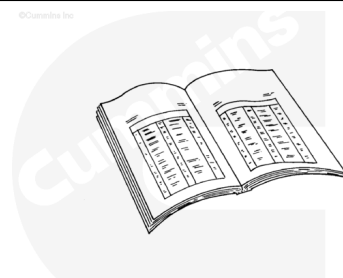
Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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Prepare the engine to remove the front crankshaft gear.

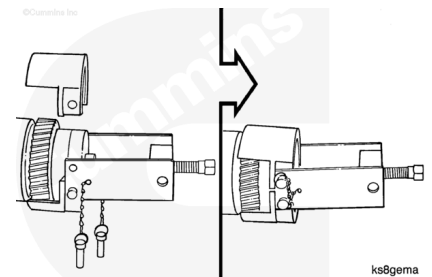
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Bar the engine to top dead center compression stroke. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-012-tr.html>)
- Remove the front gear cover and all related components. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html>)
- Remove the water pump idler gear. Refer to Procedure 001-040 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-040-tr.html>)

- Remove the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-039-tr.html)
- Remove the front gear cover spacer plate. Refer to Procedure 001-029 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-029-tr.html)

Remove

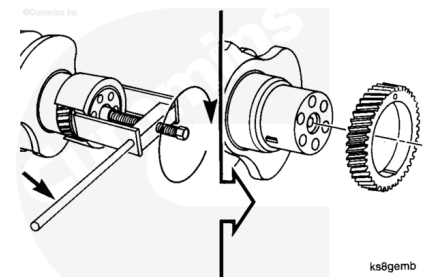
Note : Only remove the gear when the crankshaft or the gear is damaged.

Use a puller jaw, Part Number 3375835, and bridge assembly, Part Number 3375834. Install the puller jaw and bridge assembly on the gear.



⚠ CAUTION ⚠

Heat can be used to aid the removal of the gear. Do not use a cutting torch. The high temperature of a torch will damage the teeth of the gear.



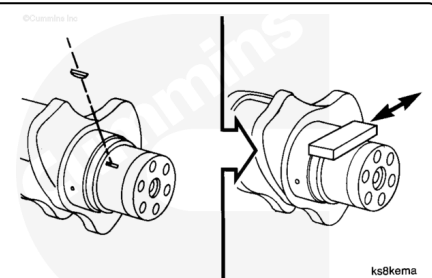
⚠ CAUTION ⚠

Do not exceed 475 N·m [350 ft-lb] of torque when turning the jackscrew or damage can occur.

Lubricate the puller jackscrew with engine oil. Hold the pry bar and turn the jackscrew to remove the gear.

Remove the key.

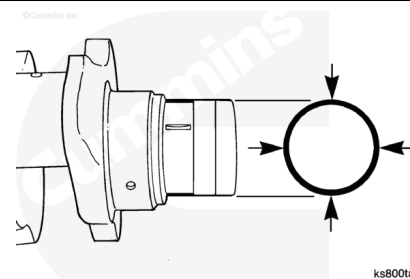
Use a hone stone and polish the shaft outside diameter. Remove all of the small scratches, burrs, and small grooves.



Inspect for Reuse

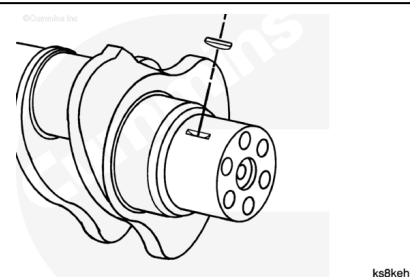
Clean and inspect the crankshaft for reuse. Measure the outside diameter

Crankshaft outside diameter (Gear Location)		
mm		in
111.25	MIN	4.380
111.28	MAX	4.381



Install

Install the key.



⚠ CAUTION ⚠

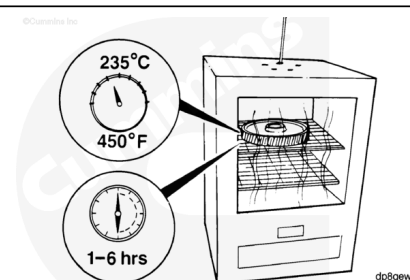
Wear protective clothing to prevent personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat.



Use an oven and adjust the heat to 235°C [455°F]. Heat the gear for a **minimum** of 1 hour, but **no** more than a maximum of 6 hours. The inside diameter of the gear will become larger

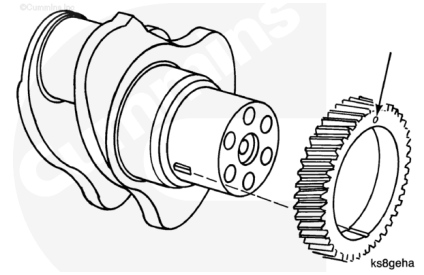
and will simplify the installation.

⚠ CAUTION ⚠

Wear protective clothing to prevent personal injury from burns.

⚠ CAUTION ⚠

Allow the gear to air cool slowly. Do not use water or oil to reduce the cooling time. This will cause the gear to crack.



Note : The timing mark on the crankshaft gear must be visible from the front of the gear after it is installed on the crankshaft.

Lubricate the outside diameter of the crankshaft with Lubriplate® Number 105 multi-purpose lubricant, or equivalent. Remove the gear from the oven and install it on the crankshaft. The keyway in the gear **must** be aligned with the key in the crankshaft.

Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Prepare the engine for normal operation.

- Install the front gear cover spacer plate. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)
- Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html>)
- Install the hydraulic pump idler gear. Refer to Procedure 001-039 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-039-tr.html)

- Install the water pump idler gear. Refer to Procedure 001-040 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-040-tr.html)

- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html)

- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

Start the engine and check for leaks.